INCH-POUND
MIL-DTL-17931F
7 February 2025
SUPERSEDING
MIL-B-17931E
27 July 1987

DETAIL SPECIFICATION
BEARINGS, BALL, ANNULAR, FOR QUIET OPERATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification provides the requirements, verification procedures, and application requirements for annular ball bearings used in quiet (low-vibration) operating machinery. This specification includes bearings with industry bore codes 00 through 22 and 32.

1.2 Classification. The bearings are to be identified by the standard part numbers designated in 1.3, as specified (see 6.2). The bearings specified herein are of the following types and classes. The suffixes and prefixes given below are included with the bearings' Navy code designation (see 1.4).

a. Type 111 – Single row, radial, non-loading groove, both rings have the same width, metric.
   (1) Class 1 – Open.
   (2) Class 2 – Single shield (suffix Z).
   (3) Class 3 – Double shield (suffix ZZ).
   (4) Class 7 – Single contact seal (suffix S) (see 6.6.1).
   (5) Class 8 – Double contact seal (suffix SS).

b. Type 120 – Single row, radial, non-loading groove, both rings have the same width, sealed (cartridge type without grease plug), metric bore and outside diameter (OD), inch width. Type 120 bearings have the prefix S3.
   (1) Class 1 – Noncontact or labyrinth seals.
   (2) Class 2 – Synthetic contact seals.

c. Type 131 – Single row, counterbore (primarily radial), contact angle: 8 degrees-30 minutes to 10 degrees-15 minutes. Self-contained, both rings have the same width, metric.
   (1) Class 1 – Single bearing (suffix R).
   (2) Class 2 – Duplex pair (suffix RD), the faces are ground for universal mounting: face-to-face (suffix RDF), back-to-back (suffix RDB), or tandem (suffix RDT).

d. Type 133 – Single row, angular contact (contact angle: 25 degrees or less), self-contained, both rings have the same width, metric. Angular contact bearings have the prefix 7.
   (1) Class 2 – Duplex pair (suffix D), the faces are ground for universal mounting: face-to-face (suffix DF), back-to-back (suffix DB), or tandem (suffix DT)

e. Type 134 – Single row, angular contact (contact angle: greater than 25 degrees, up to 45 degrees), self-contained, both rings have the same width, metric. Angular contact bearings have the prefix 7.
   (1) Class 1 – Single bearing. 7310 type 134, class 1.
   (2) Class 2 – Duplex pair (suffix D), the faces are ground for universal mounting: face-to-face (DF), back-to-back (DB), or tandem (DT). Bearings with suffix DF or DB are made for that mounting only.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 3110

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1.3 Part or identifying number (PIN). PINs to be used for annular ball bearings acquired to this specification are created as follows:

M	17931	-	XXXX
Prefix for Military Specification	Specification No.		Bearing part number

Example: M17931-G064

1.4 Navy code. The Navy code consists of a three-digit basic bearing number with a prefix, a suffix, or both, included as necessary (see 1.2). The three-digit basic bearing number identifies the external boundary dimensions. The first digit of the basic bearing number identifies the bearing series (or weight) as follows:

a. 1 - Extra-light (for all types except 120).
b. 2 - Light (for all types except 120).
c. 3 - Medium (for all types except 120).
d. 4 - Heavy (for all types except 120).
e. 5 - Light (for type 120 bearings only).
f. 6 - Medium (for type 120 bearings only).

The second and third digits of the basic bearing number are the industry bore code. When multiplied by 5, these digits give the bearing bore diameter in millimeters for bore codes 04 and greater (this rule does not apply for bore codes 00 through 03).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-6085 - Lubricating Oil, Instrument, Aircraft, Low Volatility
MIL-PRF-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service
MIL-PRF-24508 - Grease, High Performance, Multipurpose

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A295/A295M - Standard Specification for High-Carbon Anti-Friction Bearing Steel
ASTM E18 - Standard Test Methods for Rockwell Hardness of Metallic Materials
ASTM E45 - Standard Test Methods for Determining the Inclusion Content of Steel
ASTM E112 - Standard Test Methods for Determining Average Grain Size

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ASTM F2215 - Standard Specification for Balls, Bearings, Ferrous and Nonferrous for Use in Bearings, Valves, and Bearing Applications.

(Copies of these documents are available online at www.astm.org.)

AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA)

ABMA 4 - Tolerance Definitions and Gauging Practices for Ball and Roller Bearings

ANSI/ABMA 9 - Load Ratings and Fatigue Life for Ball Bearings

ABMA/ISO 10A - Metal Balls for Unground Bearings and Other Uses

ABMA 20 - Radial Bearings of Ball, Cylindrical Roller and Spherical Roller Types - Metric Design

ABMA/ISO 15242 - Rolling Bearings –Measuring Methods for Vibration

ABMA/ISO 3290 - Rolling Bearings – Balls

(Copies of these documents are available online at www.americanbearings.org/.)

SAE INTERNATIONAL

SAE-AMS6444 - Steel Bars, Forgings, and Tubing 1.45Cr (0.93-1.05C) (SAE 52100) Premium Aircraft-Quality, Consumable Electrode Vacuum Remelted

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to a first article inspection as specified in 4.2.

3.2 Materials. The materials used in the bearings shall meet the specific requirements in 3.2.1 through 3.2.7, and 3.3, so that the finished bearings and components meet the requirements in 3.5 through 3.7. The specific materials and fabrication methods used in the bearings shall be the same as those used for the first article examination and tests unless the bearing manufacturer demonstrates to the Technical Authority's (see 6.6.7) satisfaction that the change has no detrimental effect on bearing's performance.

3.2.1 Compatibility and corrosion resistance. The bearing manufacturer shall certify that the materials selected for the bearings, including the cages, shields, and seals, are compatible with and resistant to deterioration due to lubricants, preservatives, hydraulic fluid, solvents, and other substances and chemicals that may contact the bearings. The Technical Authority reserves the right to require the bearing manufacturer to demonstrate that there is no corrosion, pitting, or other signs of deterioration on the bearing components after they are tested in accordance with 4.4.7.

3.2.2 Temperature resistance. The bearing manufacturer shall certify that the materials selected for the bearings, including the cages, shields, and seals, are capable of continuous operations up to 230 °F (110 °C) as measured on the outer bearing ring (non-rotating). The temperature on the outer bearing ring shall not exceed 300 °F (148.9 °C) when the bearings are tested in accordance with 4.4.4.

3.2.3 Inner and outer ring material. Unless approved by the Technical Authority, the material for the inner and outer rings shall be type 52100 chromium-alloy steel in accordance with SAE-AMS6444, produced by vacuum induction melt and vacuum arc remelt (VIM VAR). Intentional additions of calcium or calcium alloys for deoxidation or inclusion shape control shall not be used. The material microstructure shall be uniform and spheroidized and meet the carbide microstructure requirements of ASTM A295/A295M. The material supplier shall certify that the material meets the above requirements. Requests to use alternate material shall be submitted to the Technical Authority and shall include a detailed technical justification for the proposed alternate material.

3.2.3.1 Steel form. Starting material shall be ring rolled forging for bore size 17 and higher. Bearings with bore size less than 17 shall have flat die forging.

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3.2.4 Ball material. Unless approved by the Technical Authority, the balls shall be made from type 52100 chromium-alloy steel in accordance with ASTM A295/A295M. The material microstructure shall be uniform and spheroidized. The grain size of the material shall be determined in accordance with ASTM E112. The material supplier shall certify that the material meets the above requirements. Requests to use alternate material shall be submitted to the Technical Authority and shall include a detailed technical justification for the proposed alternate material.

3.2.5 Cage material. The bearing manufacturer shall select a cage material compatible with the materials used for the lubricant, balls, and rings and meet the bearing performance requirements in this specification.

3.2.6 Shield and seal material. The materials used for the bearing shield and seals shall be compatible with the other materials.

3.2.7 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Hardness. The hardness of the finished rings and balls shall meet the values specified in 3.3.1 and 3.3.2.

3.3.1 Hardness of the inner and outer rings. The hardness of the inner and outer rings shall be not less than 58 Hardness Rockwell C (HRC) when measured in accordance with 4.4.5. The hardness shall not vary more than three HRC points when measured on any one ring.

3.3.2 Hardness of the balls. The hardness of the balls shall be 60-67 HRC. The mean hardness (see 6.6.3) of the balls shall be at least two HRC points greater than the mean hardness of the inner and outer rings independently. All hardness measurements shall be in accordance with ASTM F2215. The balls shall not be case-hardened or work-hardened.

3.4 Construction. The bearings shall be manufactured using new, unused parts. The finished bearings shall comply with the requirements of tables I, II, or III, as applicable, for the standard part number specified (see 1.2 and 6.2). In addition, the cross-sectional radii of the inner ring raceway shall not exceed 52 percent of the nominal ball diameter, and the cross-sectional radii of the outer ring raceway shall not exceed 53 percent of the nominal ball diameter.

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TABLE I. Requirements for type 111 bearings.

Standard part number M17931-	Type/class (see 1.2)	Navy code (see 1.4)	Dimensions (inches)			Radial internal clearance min/max	Shaft shoulder diameter (min) 1/	Chamfer (min) 2/	Lubricant 3/	Minimum dynamic load rating (C_r), pounds
			Bore max/min	OD max/min	Width max (see 3.4)					
G001	111/1	200	0.3937 0.3935	1.1811 1.1809	0.3543	0.0003 0.0006	0.5906	0.024	G	1,330
G002	111/8	200SS	0.3937 0.3935	1.1811 1.1809	0.3543	0.0003 0.0006	0.5906	0.024	G	1,330
G003	111/1	101	0.4724 0.4722	1.1024 1.1022	0.3150	0.0004 0.0007	0.5709	0.012	G	1,150
G004	111/2	301Z	0.4724 0.4722	1.4567 1.4565	0.4724	0.0004 0.0007	0.7087	0.039	G	2,170
G005	111/1	202	0.5906 0.5904	1.3780 1.3778	0.4331	0.0004 0.0007	0.7874	0.024	G	1,720
G006	111/2	302Z	0.5906 0.5904	1.6535 1.6533	0.5118	0.0004 0.0007	0.8268	0.039	G	2,550
G007	111/1	203	0.6693 0.6691	1.5748 1.5746	0.4724	0.0004 0.0007	0.8661	0.024	G	2,150
G008	111/2	203Z	0.6693 0.6691	1.5748 1.5746	0.4724	0.0004 0.0007	0.8661	0.024	G	2,150
G009	111/3	203ZZ	0.6693 0.6691	1.5748 1.5746	0.4724	0.0004 0.0007	0.8661	0.024	G	2,150
G010	111/8	203SS	0.6693 0.6691	1.5748 1.5746	0.4724	0.0004 0.0007	0.8661	0.024	G	2,150
G011	111/2	303Z	0.6693 0.6691	1.8504 1.8502	0.5512	0.0004 0.0007	0.9055	0.039	G	3,040
G012	111/8	303SS	0.6693 0.6691	1.8504 1.8502	0.5512	0.0004 0.0007	0.9055	0.039	G	3,040
G013	111/1	204	0.7874 0.7872	1.8504 1.8502	0.5512	0.0005 0.0008	1.0236	0.039	G	2,890
G014	111/2	204Z	0.7874 0.7872	1.8504 1.8502	0.5512	0.0005 0.0008	1.0236	0.039	G	2,890

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TABLE I. Requirements for type 111 bearings – Continued.

Standard part number M17931-	Type/class (see 1.2)	Navy code (see 1.4)	Dimensions (inches)			Radial internal clearance min/max	Shaft shoulder diameter (min) 1/	Chamfer (min) 2/	Lubricant 3/	Minimum dynamic load rating (C_r), pounds
			Bore max/min	OD max/min	Width max (see 3.4)					
G015	111/8	204SS	0.7874 0.7872	1.8504 1.8502	0.5512	0.0005 0.0008	1.0236	0.039	G	2,890
G016	111/1	304	0.7874 0.7872	2.0472 2.0470	0.5906	0.0005 0.0008	1.0630	0.043	G	3,590
G017	111/2	304Z	0.7874 0.7872	2.0472 2.0470	0.5906	0.0005 0.0008	1.0630	0.043	G	3,590
G018	111/1	205	0.9843 0.9841	2.0472 2.0470	0.5906	0.0005 0.0008	1.2205	0.039	G	3,170
G019	111/2	205Z	0.9843 0.9841	2.0472 2.0470	0.5906	0.0005 0.0008	1.2205	0.039	G	3,170
G020	111/1	305	0.9843 0.9841	2.4409 2.4407	0.6693	0.0005 0.0008	1.2598	0.043	G	4,770
G021	111/2	305Z	0.9843 0.9841	2.4409 2.4407	0.6693	0.0005 0.0008	1.2598	0.043	G	4,770
G022	111/8	305SS	0.9843 0.9841	2.4409 2.4407	0.6693	0.0005 0.0008	1.2598	0.043	G	4,770
G023	111/1	206	1.1811 1.1809	2.4409 2.4407	0.6299	0.0005 0.0009	1.4173	0.039	G	4,390
G024	111/2	206Z	1.1811 1.1809	2.4409 2.4407	0.6299	0.0005 0.0009	1.4173	0.039	G	4,390
G025	111/1	306	1.1811 1.1809	2.8346 2.8344	0.7480	0.0005 0.0009	1.4567	0.043	G	6,340
G026	111/2	306Z	1.1811 1.1809	2.8346 2.8344	0.7480	0.0005 0.0009	1.4567	0.043	G	6,340
G027	111/3	306ZZ	1.1811 1.1809	2.8346 2.8344	0.7480	0.0005 0.0009	1.4567	0.043	G	6,340

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TABLE I. Requirements for type 111 bearings – Continued.

Standard part number M17931-	Type/class (see 1.2)	Navy code (see 1.4)	Dimensions (inches)			Radial internal clearance min/max	Shaft shoulder diameter (min) 1/	Chamfer (min) 2/	Lubricant 3/	Minimum dynamic load rating (C_r), pounds
			Bore max/min	OD max/min	Width max (see 3.4)					
G028	111/8	306SS	1.1811 1.1809	2.8346 2.8344	0.7480	0.0005 0.0009	1.4567	0.043	G	6,340
G029	111/2	406Z	1.1811 1.1809	3.5433 3.5431	0.9055	0.0005 0.0009	1.4685	0.059	G	9,700
G030	111/1	207	1.3780 1.3778	2.8346 2.8344	0.6693	0.0005 0.0008	1.6535	0.043	G	5,800
G031	111/1	307	1.3780 1.3778	3.1496 3.1494	0.8268	0.0005 0.0008	1.7126	0.059	G	7,510
G032	111/2	307Z	1.3780 1.3778	3.1496 3.1494	0.8268	0.0005 0.0008	1.7126	0.059	G	7,510
G033	111/3	307ZZ	1.3780 1.3778	3.1496 3.1494	0.8268	0.0005 0.0008	1.7126	0.059	G	7,510
G034	111/8	307SS	1.3780 1.3778	3.1496 3.1494	0.8268	0.0005 0.0008	1.7126	0.059	G	7,510
G035	111/1	208	1.5748 1.5746	3.1496 3.1494	0.7087	0.0005 0.0008	1.8504	0.043	G	6,580
G035A	111/1	208SS	1.5748 1.5746	3.1496 3.1494	0.7087	0.0005 0.0008	1.8504	0.043	G	6,580
G036	111/1	308	1.5748 1.5746	3.5433 3.5431	0.9055	0.0005 0.0008	1.9094	0.059	G	9,160
G037	111/2	308Z	1.5748 1.5746	3.5433 3.5431	0.9055	0.0005 0.0008	1.9094	0.059	G	9,910
G038	111/8	109SS	1.7717 1.7715	2.9528 2.9526	0.6299	0.0005 0.0009	2.0079	0.039	G	4,830
G039	111/1	209	1.7717 1.7715	3.3465 3.3463	0.7480	0.0005 0.0009	2.0472	0.043	G	7,380

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TABLE I. Requirements for type 111 bearings – Continued.

Standard part number M17931-	Type/class (see 1.2)	Navy code (see 1.4)	Dimensions (inches)			Radial internal clearance min/max	Shaft shoulder diameter (min) 1/	Chamfer (min) 2/	Lubricant 3/	Minimum dynamic load rating (C_r), pounds
			Bore max/min	OD max/min	Width max (see 3.4)					
G040	111/1	309	1.7717 1.7715	3.9370 3.9368	0.9843	0.0005 0.0009	2.1063	0.059	G	11,910
G041	111/2	309Z	1.7717 1.7715	3.9370 3.9368	0.9843	0.0005 0.0009	2.1063	0.059	G	11,910
G042	111/7	309S	1.7717 1.7715	3.9370 3.9368	0.9843	0.0005 0.0009	2.1063	0.059	G	11,910
G043	111/8	309SS	1.7717 1.7715	3.9370 3.9368	0.9843	0.0005 0.0009	2.1063	0.059	G	11,910
G044	111/1	210	1.9685 1.9683	3.5433 3.5431	0.7874	0.0005 0.0009	2.2441	0.043	G	7,920
G045	111/1	310	1.9685 1.9683	4.3307 4.3305	1.0630	0.0005 0.0009	2.3622	0.079	G	13,950
G046	111/2	310Z	1.9685 1.9683	4.3307 4.3305	1.0630	0.0005 0.0009	2.3622	0.079	G	13,950
G047	111/3	310ZZ	1.9685 1.9683	4.3307 4.3305	1.0630	0.0005 0.0009	2.3622	0.079	G	13,950
G048	111/8	310SS	1.9685 1.9683	4.3307 4.3305	1.0630	0.0005 0.0009	2.3622	0.079	G	13,950
G049	111/2	410Z	1.9685 1.9683	5.1181 5.1179	1.2205	0.0005 0.0009	2.3819	0.083	G	19,640
G050	111/1	211	2.1654 2.1652	3.9370 3.9368	0.8268	0.0007 0.0011	2.5000	0.059	G	9,790
G051	111/1	311	2.1654 2.1652	4.7244 4.7242	1.1417	0.0007 0.0011	2.5591	0.079	G	16,080
G052	111/2	311Z	2.1654 2.1652	4.7244 4.7242	1.1417	0.0007 0.0011	2.5591	0.079	G	16,130

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TABLE I. Requirements for type 111 bearings – Continued.

Standard part number M17931-	Type/class (see 1.2)	Navy code (see 1.4)	Dimensions (inches)			Radial internal clearance min/max	Shaft shoulder diameter (min) 1/	Chamfer (min) 2/	Lubricant 3/	Minimum dynamic load rating (C_r), pounds
			Bore max/min	OD max/min	Width max (see 3.4)					
G053	111/3	311ZZ	2.1654 2.1652	4.7244 4.7242	1.1417	0.0007 0.0011	2.5591	0.079	G	16,130
G054	111/8	311SS	2.1654 2.1652	4.7244 4.7242	1.1417	0.0007 0.0011	2.5591	0.079	G	16,130
G055	111/1	212	2.3622 2.3620	4.3307 4.3305	0.8661	0.0007 0.0011	2.6969	0.059	G	11,840
G056	111/2	212Z	2.3622 2.3620	4.3307 4.3305	0.8661	0.0007 0.0011	2.6969	0.059	G	11,840
G057	111/1	312	2.3622 2.3620	5.1181 5.1179	1.2205	0.0007 0.0011	2.8346	0.083	G	18,390
G058	111/2	312Z	2.3622 2.3620	5.1181 5.1179	1.2205	0.0007 0.0011	2.8346	0.083	G	18,460
G059	111/3	312ZZ	2.3622 2.3620	5.1181 5.1179	1.2205	0.0007 0.0011	2.8346	0.083	G	18,460
G060	111/3	213ZZ	2.5591 2.5589	4.7244 4.7242	0.9055	0.0007 0.0011	2.8937	0.059	G	12,920
G061	111/1	313	2.5591 2.5589	5.5118 5.5116	1.2992	0.0007 0.0011	3.0315	0.083	G	20,840
G062	111/2	313Z	2.5591 2.5589	5.5118 5.5116	1.2992	0.0007 0.0011	3.0315	0.083	G	20,840
G063	111/1	313	2.5591 2.5589	5.5118 5.5116	1.2992	0.0009 0.0017	3.0315	0.083	L	20,920
G064	111/1	314	2.7559 2.7557	5.9055 5.9053	1.3780	0.0008 0.0012	3.2283	0.083	G	23,520
G065	111/2	314Z	2.7559 2.7557	5.9055 5.9053	1.3780	0.0008 0.0012	3.2283	0.083	G	23,520
G066	111/3	314ZZ	2.7559 2.7557	5.9055 5.9053	1.3780	0.0008 0.0012	3.2283	0.083	G	23,520

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TABLE I. Requirements for type 111 bearings – Continued.

Standard part number M17931-	Type/class (see 1.2)	Navy code (see 1.4)	Dimensions (inches)			Radial internal clearance min/max	Shaft shoulder diameter (min) 1/	Chamfer (min) 2/	Lubricant 3/	Minimum dynamic load rating (C_r), pounds
			Bore max/min	OD max/min	Width max (see 3.4)					
G067	111/1	215	2.9528 2.9526	5.1181 5.1179	0.9843	0.0008 0.0012	3.2874	0.059	G	14,920
G068	111/1	315	2.9528 2.9526	6.2992 6.2990	1.4567	0.0008 0.0012	3.4252	0.083	G	25,620
G069	111/2	315Z	2.9528 2.9526	6.2992 6.2990	1.4567	0.0008 0.0012	3.4252	0.083	G	25,620
G070	111/2	315Z	2.9528 2.9526	6.2992 6.2990	1.4567	0.0008 0.0012	3.4252	0.083	L	25,620
G071	111/3	216ZZ	3.1496 3.1493	5.5118 5.5116	1.0236	0.0008 0.0012	3.5433	0.079	G	16,390
G072	111/1	316	3.1496 3.1493	6.6929 6.6926	1.5354	0.0008 0.0012	3.6220	0.083	G	27,770
G073	111/2	316Z	3.1496 3.1493	6.6929 6.6926	1.5354	0.0008 0.0012	3.6220	0.083	G	27,770
G074	111/1	217	3.3465 3.3462	5.9055 5.9053	1.1024	0.0009 0.0014	3.7402	0.079	G	18,780
G075	111/3	217ZZ	3.3465 3.3462	5.9055 5.9053	1.1024	0.0009 0.0014	3.7402	0.079	G	18,780
G076	111/1	317	3.3465 3.3462	7.0866 7.0863	1.6142	0.0009 0.0014	3.8976	0.118	G	29,970
G077	111/2	317Z	3.3465 3.3462	7.0866 7.0863	1.6142	0.0009 0.0014	3.8976	0.118	G	29,970
G078	111/1	318	3.5433 3.5430	7.4803 7.4800	1.6929	0.0009 0.0014	4.0945	0.118	G	32,210
G078A	111/1	219	3.7402 3.7399	6.6929 6.6926	1.2598	0.0009 0.0014	4.1733	0.083	G	24,500
G079	111/1	319	3.7402 3.7399	7.8740 7.8737	1.7717	0.0009 0.0014	4.2913	0.118	G	34,500

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TABLE I. Requirements for type 111 bearings – Continued.

Standard part number M17931-	Type/class (see 1.2)	Navy code (see 1.4)	Dimensions (inches)			Radial internal clearance min/max	Shaft shoulder diameter (min) 1/	Chamfer (min) 2/	Lubricant 3/	Minimum dynamic load rating (C_r), pounds
			Bore max/min	OD max/min	Width max (see 3.4)					
G080	111/2	319Z	3.7402 3.7399	7.8740 7.8737	1.7717	0.0009 0.0014	4.2913	0.118	G	34,500
G081	111/2	220Z	3.9370 3.9367	7.0866 7.0863	1.3386	0.0009 0.0014	4.4094	0.083	G	27,580
G082	111/2	320Z	3.9370 3.9367	8.4646 8.4643	1.8504	0.0009 0.0014	4.4882	0.118	G	39,070
G083	111/1	320	3.9370 3.9367	8.4646 8.4643	1.8504	0.0018 0.0023	4.4882	0.118	G	38,950
G084	111/1	321	4.1339 4.1336	8.8583 8.8580	1.9291	0.0011 0.0016	4.6850	0.118	G	41,490
G085	111/1	222	4.3307 4.3304	7.8740 7.8737	1.4961	0.0011 0.0016	4.8031	0.083	G	32,420
G086	111/1	322	4.3307 4.3304	9.4488 9.4485	1.9685	0.0011 0.0016	4.8819	0.118	G	48,190
G087	111/1	322	4.3307 4.3304	9.4488 9.4485	1.9685 4/	0.0024 0.0030	4.8819	0.118	G	46,100
G088	111/1	232	6.2992 6.2988	11.4173 11.4170	1.8898	0.0013 0.0021	6.8504	0.118	G	49,990

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TABLE I. Requirements for type 111 bearings – Continued.

NOTES:

1/ The shaft shoulder diameter is for reference only. The maximum shoulder heights shall permit the bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, other provisions shall be made so the bearing can be removed from the shaft without damaging the bearing or the other parts.

2/ The minimum bearing chamfer shall clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.

3/ L lubricant shall be oil in accordance with MIL-PRF-17331; preservative shall be in accordance with MIL-PRF-6085.
G lubricant/preservative shall be grease in accordance with MIL-PRF-24508. Selection of lubricant shall be in accordance with A.3.2. and is based on actual equipment operating conditions.

4/ Under 15-pound axial gage load, the unloaded side face of the inner ring shall stick out 0.0110 – 0.0115 inches beyond the outer ring side face for both loading directions. Inner ring width variance to satisfy the stickout requirement is permitted.

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TABLE II. Requirements for type 120 bearings.

Standard part number M17931-	Type/class (see 1.2)	Navy code (see 1.4)	Dimensions (inches)			Radial internal clearance min/max	Shaft shoulder diameter (min) 1/	Chamfer (min) 2/	Lubricant 3/	Minimum dynamic load rating (C_r), pounds
			Bore max/min	OD max/min	Width max (see 3.4)					
C001	120/2	S3502	0.5906 0.5904	1.3780 1.3778	0.6250	0.0004 0.0007	0.07874	0.024	G	1,720
C002	120/2	S3503	0.6693 0.6691	1.5748 1.5746	0.6875	0.0004 0.0007	0.8661	0.024	G	2,240
C003	120/2	S3504	0.7874 0.7872	1.8504 1.8502	0.8125	0.0005 0.0008	1.0236	0.039	G	2,890
C004	120/1	S3604	0.7874 0.7872	2.0472 2.0470	0.8750	0.0005 0.0008	1.0630	0.043	G	3,590
C005	120/1	S3505	0.9843 0.9841	2.0472 2.0470	0.8125	0.0005 0.0008	1.2205	0.039	G	3,170
C006	120/2	S3605	0.9843 0.9841	2.4409 2.4407	1.0000	0.0005 0.0008	1.2598	0.043	G	4,770
C007	120/1	S3506	1.1811 1.1809	2.4409 2.4407	0.9375	0.0005 0.0008	1.4173	0.039	G	4,390
C008	120/2	S3606	1.1811 1.1809	2.8346 2.8344	1.1875	0.0005 0.0008	1.4567	0.043	G	6,340
C009	120/2	S3507	1.3780 1.3778	2.8346 2.8344	1.0625	0.0005 0.0008	1.6335	0.043	G	5,800
C010	120/2	S3607	1.3780 1.3778	3.1496 3.1494	1.3750	0.0005 0.0008	1.7126	0.059	G	7,510
C011	120/1 2/	S3607	1.3780 1.3778	3.1496 3.1494	1.3750	0.0005 0.0008	1.7126	0.059	G	7,510
C012	120/1	S3608	1.5748 1.5746	3.5433 3.5431	1.4375	0.0005 0.0008	1.9094	0.059	G	9,910
C013	120/1	S3509	1.7717 1.7715	3.3465 3.3463	1.1875	0.0005 0.0009	2.0472	0.043	G	7,380
C014	120/1	S3609	1.7717 1.7715	3.9370 3.9368	1.5625	0.0005 0.0009	2.0163	0.059	G	11,910

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TABLE II. Requirements for type 120 bearings – Continued.

Standard part number M17931-	Type/class (see 1.2)	Navy code (see 1.4)	Dimensions (inches)			Radial internal clearance min/max	Shaft shoulder diameter (min) 1/	Chamfer (min) 2/	Lubricant 3/	Minimum dynamic load rating (C_r), pounds
			Bore max/min	OD max/min	Width max (see 3.4)					
C015	120/2	S3610	1.9685 1.9683	4.3307 4.3305	1.7500	0.0005 0.0009	2.3622	0.079	G	13,950
C016	120/2 2/	S3610	1.9685 1.9683	4.3307 4.3305	1.7500	0.0005 0.0009	2.3622	0.079	G	13,950
C017	120/1	S3611	2.1654 2.1652	4.7244 4.7242	1.9375	0.0007 0.0011	2.5591	0.079	G	16,130
C018	120/2	S3612	2.3622 2.3620	5.1181 5.1179	2.1250	0.0007 0.0011	2.8346	0.079	G	18,460
C019	120/2	S3613	2.5591 2.5589	5.5118 5.5116	2.3125	0.0007 0.0011	3.0315	0.083	G	20,800
C020	120/1	S3614	2.7559 2.7557	5.9055 5.9053	2.5000	0.0008 0.0012	3.2283	0.083	G	23,400
C021	120/1	S3615	2.9528 2.9526	6.2992 6.2990	2.6875	0.0008 0.0012	3.4252	0.083	G	25,610
C022	120/2	S3618	3.5433 3.5430	7.4803 7.4800	2.8750	0.0009 0.0014	4.0945	0.083	G	32,110

NOTES:

1/ The shaft shoulder diameter is for reference only. The maximum shoulder heights shall permit the bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, other provisions shall be made so the bearing can be removed from the shaft without damaging the bearing or the other parts.

2/ The minimum bearing chamfer shall clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.

3/ G lubricant/preservative shall be grease in accordance with MIL-PRF-24508. Selection of lubricant shall be in accordance with A.3.2 and is based on actual equipment operating conditions.

4/ One seal is removed.

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TABLE III. Requirements for type 131, 133, and 134 bearings.

Standard part number M17931-	Type/class (see 1.2)	Navy code (see 1.4)	Dimensions (inches)			Shoulder diameter [2/]	Chamfer [3/]	Lubricant [4/]	Preload (pounds) [5/]	Negative stickout max/min (inches) [6/] (see 6.6.4)	Minimum dynamic load rating (C_r), pounds
			Bore max/min	OD max/min	Width max [1/] (see 3.4)						
A001	134/2	7203D	0.6693 0.6691	1.5748 1.5746	0.9448	0.8661	0.0236	G	20	--	2,370
A002	134/2	7303D	0.6693 0.6691	1.8504 1.8502	1.1024	0.9055	0.0394	G	20	--	3,690
A003	134/2	7204D	0.7874 0.7872	1.8504 1.8502	1.1024	1.0236	0.0394	G	100	--	3,280
A004	134/2	7305D	0.9843 0.9841	2.4409 2.4407	1.3386	1.2598	0.0433	G	50	--	6,440
A005	134/2	7306D	1.1811 1.1809	2.8346 2.8344	1.4960	1.4567	0.0433	G	50	--	5,950
A006	134/2	7207D	1.3780 1.3778	2.8346 2.8344	1.3386	1.6535	0.0433	G	--	0.0001 0.0003	4,930
A007	134/2	7207D	1.3780 1.3778	2.8346 2.8344	1.3386	1.6535	0.0433	G	50	--	6,540
A008	134/2	7307D	1.3780 1.3778	3.1496 3.1494	1.6536	1.7126	0.0591	G	50	--	9,630
A009	134/2	7407D	1.3780 1.3778	3.9370 3.9368	1.9686	1.7323	0.0591	G	50	--	13,700
A010	134/2	7308D	1.5748 1.5746	3.5433 3.5431	1.8110	1.9094	0.0591	G	--	0.0003 0.0005	11,750
A011	134/2	7308D	1.5748 1.5746	3.5433 3.5431	1.8110	1.9094	0.0591	G	50	--	9,370
A011A	134/2	7308D	1.5748 1.5746	3.5433 3.5431	1.8110	1.9094	0.0591	G	150	--	9,370
A012	134/2	7408D	1.5748 1.5746	4.3307 4.3305	2.1260	1.9685	0.0787	G	50	--	14,800

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TABLE III. Requirements for type 131, 133, and 134 bearings – Continued.

Standard part number M17931-	Type/class (see 1.2)	Navy code (see 1.4)	Dimensions (inches)			Shoulder diameter 2/	Chamfer 3/	Lubricant 4/	Preload (pounds) 5/	Negative stickout max/min (inches) 6/ (see 6.6.4)	Minimum dynamic load rating (C_r), pounds
			Bore max/min	OD max/min	Width max 1/ (see 3.4)						
A013	134/2	7209D	1.7717 1.7715	3.3465 3.3463	1.4960	2.0472	0.0433	G	50	--	6,470
A014	134/2	7309D	1.7717 1.7715	3.9370 3.9368	1.9686	2.1062	0.0591	G	50	--	13,870
A015	134/2	7309D	1.7717 1.7715	3.9370 3.9368	1.9686	2.1062	0.0591	G	100	--	13,870
A016	134/1	7310	1.9685 1.9683	4.3307 4.3305	1.0630	2.3622	0.0787	G	--	--	16,250
A017	134/2	7310D	1.9685 1.9683	4.3307 4.3305	2.1260	2.3622	0.0787	G	100	--	15,710
A017A	134/2	7310D	1.9685 1.9683	4.3307 4.3305	2.1260	2.3622	0.0787	G	350	--	15,710
A017B	134/2	7310D	1.9685 1.9683	4.3307 4.3305	2.1260	2.3622	0.0787	G	600	--	15,710
A018	134/2	7310D	1.9685 1.9683	4.3307 4.3305	2.1260	2.3622	0.0787	G	--	0.0003 0.0005	15,710
A019	134/2	7410D	1.9685 1.9683	5.1181 5.1179	2.4410	2.4409	0.0827	G	--	0.0003 0.0005	21,810
A020	134/2	7211D	2.1654 2.1652	3.9370 3.9368	1.6536	2.5000	0.0591	G	--	0.0004 0.0006	10,260
A021	134/2	7211D	2.1654 2.1652	3.9370 3.9368	1.6536	2.5000	0.0591	G	100	--	11,340
A022	134/2	7311D	2.1654 2.1652	4.7244 4.7242	2.2834	2.559	0.0787	G	--	0.0004 0.0006	18,150

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TABLE III. Requirements for type 131, 133, and 134 bearings – Continued.

Standard part number M17931-	Type/class (see 1.2)	Navy code (see 1.4)	Dimensions (inches)			Shoulder diameter [2/]	Chamfer [3/]	Lubricant [4/]	Preload (pounds) [5/]	Negative stickout max/min (inches) [6/] (see 6.6.4)	Minimum dynamic load rating (C_r), pounds
			Bore max/min	OD max/min	Width max [1/] (see 3.4)						
A023 [9/]	134/2	7311D	2.1654 2.1652	4.7244 4.7242	2.2834	2.559	0.0787	G	200	--	18,200
A024	134/2	7312DB	2.3622 2.3620	5.1181 5.1179	2.4410	2.8346	0.0827	G	--	0.0004 0.0006	22,750
A025	134/2	7312DF	2.3622 2.3620	5.1181 5.1179	2.4410	2.8346	0.0827	G	0	0±0.0001	20,800
A026 [9/]	134/2	7312D	2.3622 2.3620	5.1181 5.1179	2.4410	2.8346	0.0827	G	100	--	20,800
A027	134/2	7412D	2.3622 2.3620	5.9055 5.9053	2.7560	2.8346	0.0827	G	100	--	27,250
A028	134/2	7313D	2.5591 2.5589	5.5118 5.5116	2.5984	3.0315	0.0827	L	--	0.0005 0.0007	22,980
A029	134/2	7313D	2.5591 2.5589	5.5118 5.5116	2.5984	3.0315	0.0827	G	100	--	22,980
A030	134/2	7215D	2.9528 2.9526	5.1181 5.1179	1.9686	3.2874	0.0591	G	100	--	16,610
A031 [9/]	134/2	7315D	2.9528 2.9526	6.2992 6.2990	2.9134	3.4252	0.0827	L	--	0.0009 0.0011	28,820
A032	134/2	7315D	2.9528 2.9526	6.2992 6.2990	2.9134	3.4252	0.0827	G	100	--	29,220
A033	134/2	7217D	3.3465 3.3462	5.9055 5.9053	2.2048	3.7402	0.0787	G	100	--	19,300
A034	134/2	7317D	3.3465 3.3462	7.0866 7.0863	3.2284	3.8976	0.1181	L	--	0.0014 0.0016	34,460

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TABLE III. Requirements for type 131, 133, and 134 bearings – Continued.

Standard part number M17931-	Type/class (see 1.2)	Navy code (see 1.4)	Dimensions (inches)			Shoulder diameter [2/]	Chamfer [3/]	Lubricant [4/]	Preload (pounds) [5/]	Negative stickout max/min (inches) [6/] (see 6.6.4)	Minimum dynamic load rating (C_r), pounds
			Bore max/min	OD max/min	Width max [1/] (see 3.4)						
A035	134/2	7317D	3.3465 3.3462	7.0866 7.0863	3.2284	3.8976	0.1181	G	100	--	34,460
A036	131/1	318R	3.5433 3.5430	7.4803 7.4800	1.6929	4.0945	0.1181	G	--	--	35,380
A037	131/2	318RD	3.5433 3.5430	7.4803 7.4800	3.3858	4.0945	0.1181	G	500	--	35,380
A038	133/2	7318D	3.5433 3.5430	7.4803 7.4800	3.3858	4.0945	0.1181	G	500	--	35,380
A039 [9/]	134/2	7319D	3.7402 3.7399	7.8740 7.8737	3.5434	4.2913	0.1181	G	--	0.0001 [8/] 0.0003	41,070
A040	134/2	7319D	3.7402 3.7399	7.8740 7.8737	3.5434	4.2913	0.1181	G	100	--	42,330
A041	134/2	7319D	3.7402 3.7399	7.8740 7.8737	3.5434	4.2913	0.1181	G	600	--	42,330
A042 [9/]	134/2	7319D	3.7402 3.7399	7.8740 7.8737	3.5434	4.2913	0.1181	G	1,000	--	41,070
A043	133/2	7221D	4.1339 4.1336	7.4803 7.4800	2.8346	4.6063	0.0827	G	200	--	31,360

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TABLE III. Requirements for type 131, 133, and 134 bearings – Continued.

Standard part number M17931-	Type/class (see 1.2)	Navy code (see 1.4)	Dimensions (inches)			Shoulder diameter 2/	Chamfer 3/	Lubricant 4/	Preload (pounds) 5/	Negative stickout max/min (inches) 6/ (see 6.6.4)	Minimum dynamic load rating (C_r), pounds
			Bore max/min	OD max/min	Width max 1/ (see 3.4)						
A044 9/	134/2	7322DB	4.3307 4.3304	9.4488 9.4485	3.9370	4.8819	0.1181	G	2,400	--	52,190
A044A 9/,10/	134/2	7322DB	4.3307 4.3304	9.4488 9.4485	3.9370	4.8819	0.1181	G	2,400	--	52,190
A045	134/2	7322DB	4.3307 4.3304	9.4488 9.4485	3.9370	4.8819	0.1181	G	2,400	--	53,900
A045A 9/,10/	134/2	7322DB	4.3307 4.3304	9.4488 9.4485	3.9370	4.8819	0.1181	G	2,400	--	53,900

NOTES:

1/ The maximum width for individual bearings of duplex pairs is one-half the listed value.

2/ The shaft shoulder diameter is for reference only. The maximum shoulder heights shall permit the bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, other provisions shall be made so the bearing can be removed from the shaft without damaging the bearing or other parts.

3/ The minimum bearing chamfer shall clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.

4/ L lubricant shall be oil in accordance with MIL-PRF-17331; preservative shall be in accordance with MIL-PRF-6085.
G lubricant/preservative shall be grease in accordance with MIL-PRF-24508. Selection of lubricant shall be in accordance with A.3.2 and is based on actual equipment operating conditions.

5/ Preload tolerance: ±10 percent.

6/ Negative stickout: 2× the negative stickout corresponds to the unmounted axial end play of the clamped-up duplex pair.

7/ The dynamic load ratings for duplex bearings listed in this table are determined for a single bearing. Appendix A requires that duplex bearings shall be treated as two individual bearings.

8/ Stickout determined at 15 pounds thrust.

9/ Requirements of table I apply to this bearing.

10/ Requirements of table V (bore code 22SB) apply to this bearing.

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3.4.1 Types 111 and 120. Bearing types 111 and 120 shall be designed and constructed primarily for radial loads but shall be capable of supporting thrust loads from either direction. These bearing types shall conform to the requirements in tables I (type 111) and II (type 120).

3.4.2 Type 131, class 1; type 133, class 1; and type 134, class 1. Bearing types 131, 133, and 134 shall support thrust loads in one direction in combination with radial loads. These bearing types shall conform to the requirements of table III. The bearings are self-contained for handling. One raceway shoulder of the outer ring shall be cut away to permit side assembly of the inner ring and balls into the outer ring without damage to the component parts while retaining all the parts in proper operating position during normal handling.

3.4.2.1 Type 134, class 1. The manufacturer-specified unmounted nominal contact angles shall be greater than 25 degrees, up to a maximum of 45 degrees.

3.4.3 Type 131, class 2; type 133, class 2; and type 134, class 2. The construction of individual bearings of the duplex pairs shall conform to the requirements of 3.4.2 and table III. Except for DB pairs (see 3.4.8), the bearing ring side faces shall be universal ground for the specified internal condition (see 3.4.9). The pairs shall be matched by the bearing manufacturer and the matched pairs shall be packaged together.

3.4.4 Critical service. Critical service bearings are identified in table IV. All critical service bearings shall comply with the additional design details specified in table IV in addition to the requirements specified in tables I, II, or III. Alternate configurations of bearings listed in table IV shall be submitted to the technical authority for approval.

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TABLE IV. Supplemental design requirements for critical bearings.

Standard part number, M17931-	Type/class (see 1.2)	Navy code (see 1.4)	Nominal contact angle (degrees) 1/	Minimum nominal ball diameter (inch)	Minimum ball complement
G036	111/1	308	N/A	0.5938	8
G051	111/1	311	N/A	0.8125	8
G057	111/1	312	N/A	0.8750	8
G061	111/1	313	NA	0.9375	8
G062	111/2	313Z	N/A	0.9375	8
G083	111/1	320	N/A	1.4375	8
G085	111/1	222	N/A	1.1250	10
G087	111/1	322	N/A	1.6250	8
A023	134/2	7311D	30	0.8125	11
A025	134/2	7312DF	30	0.8750	11
A026	134/2	7312D	30	0.8750	11
A031	134/2	7315D	30	1.0625	11
A039	134/2	7319D	30	1.3125	12
A042	134/2	7319D	30	1.3125	12
A044	134/2	7322DB	30	1.6250	11
A044A 2/	134/2	7322DB	30	1.6250	11
A045	134/2	7322DB	25	1.5000	13
A045A 2/	134/2	7322DB	25	1.5000	13

NOTE:

1/ Contact angle tolerance shall be ±1°.

2/ Requirements of table V (bore code 22SB) apply to this bearing.

3.4.5 Balls. The balls used in the same bearing shall be selected from the same unit container. Bearing balls for all types and classes shall be in accordance with ABMA/ISO 3290 and the other requirements as specified herein. The bearing manufacturer shall select the ball grade to meet the dynamic load rating and vibration performance requirements in 3.5 and 3.6, respectively.

3.4.6 Cage. The cage shall be free of cracks, loose particles or fragments. Cage pins if utilized shall not protrude beyond the cage surface. Phenolic cage grease ports, if applicable, shall be evenly spaced and located at the center of the cage pockets to avoid possible detrimental effect on structural integrity of the cage.

3.4.6.1 Phenolic cage requirements. The following requirements are for ball riding and inner ring riding phenolic cages used in standard part numbers M17931-A044 and M17931-G085 only. Cages that are constructed from cotton fabric reinforced phenolic shall be post-cure stabilized against dimensional change and oil presoaked in MIL-PRF-24508 base oil. The finished cages shall be stored in a dry, sealed bag before installation in a bearing. Phenolic cages shall have the following process requirements. The cage vendor shall submit test data obtained from one sample ring made from each lot of raw material showing that cage dimensional stability results from post-cure temperature cycling of the phenolic material prior to final machining. Test data shall provide inner diameter measurement at three locations after each temperature cycle. One cycle is defined as 8 hours at 250 °F followed by 16 hours at ambient temperature. Testing shall be done in a dry environment. The cage is considered dimensionally stable when the inner diameter measurement variation is 0.0005 inch or less over three sequential temperature cycles. Typical dimensional stability is reached within 15 cycles.

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3.4.7 Shields and seals. An installed shield shall have a small gap between the cap inner diameter and the inner-ring and an installed seal shall have rubbing contact with the inner ring. Shields and seals, when utilized, shall not affect the dimensional tolerances specified herein. Contact seals (see 6.6.1) shall not inhibit rotation of the inner ring. Shields and seals shall not shed loose material or show evidence of concavity, convexity, waviness, denting, scratches or other surface irregularities. When a single shield or seal is required for a bearing utilizing a snap-in cage, the seal or shield shall be installed on the same side of the bearing as the closed side of the cage and on the opposite side to the marked side of the bearing.

3.4.8 Metrology. The finished bearings shall meet the metrology requirements specified in tables I, II, III, and IV, as applicable. The width tolerances of single bearings shall be plus 0.000, minus 0.002 inch. For duplex bearings, the total width tolerance of the inner and outer rings shall be plus 0.000, minus 0.010 inch. Subsequent to establishing the bearing internal condition (see 3.4.9), DB pair outer ring non-thrust faces shall be ground to a total width tolerance of minus 0.008 to minus 0.010 inch. All other bearing metrology requirements, such as ring and raceway runouts and parallelism, shall be in accordance with tolerance class ABEC-7 of ABMA 20.

3.4.9 Negative stickout or preload of duplex pairs. Type 131, class 2; type 133, class 2; and type 134, class 2 bearings shall be manufactured to the limits of negative stickout (see 6.6.4), stated in inches, or of axial preload, stated in pounds, as listed in table III. For negative stickout duplex pairs, under the minimum practical thrust gauging load used to establish free-end play, the inner and outer ring side faces of individual bearings shall lie in parallel planes within the limits specified in table III for negative stickout. For preloaded duplex pairs (see 6.6.6), a clamping force equal to the preload value of table III shall be used to bring the inner and outer ring side faces into flush contact.

3.5 Dynamic radial load rating and fatigue life. When specified (see 6.2), the bearing manufacturer shall certify that the bearings meet the basic dynamic radial load ratings (C_r, as defined in ANSI/ABMA 9) specified in tables I, II, and III. For load rating calculations, the manufacturer shall assume a zero-contact angle for type 111 (see table I) and type 120 (see table II) bearings. A fatigue test shall be conducted during the first article inspection on a sample of bearings in accordance with 4.4.4. The Technical Authority also reserves the right to require the bearing manufacturer to perform the fatigue life test on a sample of bearings offered for delivery.

3.5.1 Acceptance criteria for the dynamic radial load, fatigue life test. The temperature of the outer ring of the bearing shall not exceed 230 °F (110 °C) for the duration of the fatigue test. The bearings shall meet the following criteria when they are disassembled and inspected after the test. The following defects are only significant when they are visible to the unaided eye (corrected to 20/20 vision, as necessary). Magnification may be used to identify the nature of particular damage, provided the damage is visible to the unaided eye.

a. There shall be no breaks or cracks in the balls, rings, cage, housing, or other components.

b. There shall be no surface pits, spalls, or other indications of surface distress in the balls or raceways.

3.6 Vibration. The vibration level of individual bearings and the average vibration level of bearing lot samples shall meet the requirements specified in table V for the low-, middle-, and high-frequency ranges when tested as specified in 4.4.3. Duplex pairs shall be tested and evaluated as single bearings and not as pairs.

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TABLE V. Allowable vibration limits for individual bearings and bearing lots.

	Maximum allowable vibration for individual bearings (anderons)			Maximum allowable average vibration for bearing lots (anderons)		
Bearing	**Frequency band**			**Frequency band**		
Industry bore code	**Low**	**Middle**	**High**	**Low**	**Middle**	**High**
00	16	12	10	12	10	10
01	16	12	10	12	10	10
02	20	16	10	14	12	10
03	20	16	16	15	12	10
04	20	16	16	16	12	12
05	24	17	16	17	13	12
06	24	18	16	18	14	12
07	24	19	20	20	15	16
08	24	20	20	20	16	16
09	31	20	21	23	16	17
10	32	21	22	24	17	18
11	32	22	23	24	18	19
12	32	23	24	24	19	20
13	32	23	24	24	19	20
14	38	24	25	30	20	21
15	40	25	26	32	21	22
16	40	26	27	32	22	23
17	40	26	28	32	22	24
18	40	27	29	32	23	25
19	40	28	30	32	24	26
20	40	29	31	40	25	27
21	49	30	32	40	26	28
22	50	30	33	40	26	29
22SB	40	26	26	NA	NA	NA
24	50	31	34	40	27	30
26	57	32	36	49	28	32
32	68	40	40	60	34	36

3.7 Lubricant. The lubricants specified in tables I, II, or III shall be used with the bearings. Grease-lubricated bearings shall be preserved and pre-lubricated in accordance with MIL-PRF-24508. Oil-lubricated bearings shall be preserved in accordance with MIL-PRF-6085. When grease is specified, the bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 30- to 50-percent filled with grease.

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3.8 Marking. A die stamp shall not be used to mark the bearings. Each bearing shall be permanently marked as follows:

a. The complete standard part number (see 1.3), the manufacturer's name or symbol, and the complete bearing number shall be marked where space is available.

b. A serial number shall be marked on the outer ring face or back. For duplex pairs, the same serial number shall be used for each bearing, designating one bearing as XXXA and the other as XXXB.

c. The direction of thrust load for all classes of types 131, 133, and 134 bearings shall be preferably on the outer ring face of the bearing. If there is not sufficient space on the outer ring face, then the direction of thrust shall be marked on the inner ring face, located on the same side as the counter bored outer ring.

d. The high point of radial runout on both inner and outer rings of duplex pairs shall be marked.

e. The design preload in pounds (including "0") for all duplex pairs shall be marked.

f. The symbol "NT5" shall be marked on the outer ring.

3.9 Interchangeability. All complete bearings with the same standard part number (see 1.3) shall be functionally and dimensionally interchangeable.

3.10 Workmanship. The bearing functional (see 6.6.3) and nonfunctional (see 6.6.6) surfaces shall not be contaminated by foreign particles, mill scale, or dirt and shall be free of corrosion. There shall be no mechanical or physical damage to the functional surfaces (see 6.6.2) of the rings or balls that is detrimental to the bearing performance. Such damage includes but is not limited to tool marks, grinding scratches, pits, indentations, and cracks visible to the unaided eye (corrected to 20/20 vision, as necessary). The surface of the bearings shall be uniform in appearance and free of visible defects.

3.11 Application requirements. Equipment designers and manufacturers shall comply with appendix A.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When specified (see 6.2), samples of the bearing materials and assembled bearings of part number M17931-G061 shall be subjected to the first article inspection and tests specified in table VI. All the tested materials and bearings shall meet the performance requirements specified in table VI in order to satisfy the first article requirements. The material and bearing defects shall be classified in accordance with table VII.

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TABLE VI. First article examination and tests.

Examination or test [1/]	Test specimen	Performance requirement	Verification procedure
Properties of ring material [2/]	Sample of ring material	3.2.3	4.4.6.1
Properties of ball material [2/]	Sample of ball material	3.2.4	4.4.6.2
Vibration	32 finished bearings	3.6	4.4.3
Workmanship	32 finished bearings	3.10	4.4.2
Boundary dimensions	32 finished bearings	3.4.7, 3.4.8	4.4.1.1
Radial internal clearance	32 finished bearings	3.4.7	4.4.1.1
Basic dynamic radial load and fatigue life	20 finished bearings	3.6	4.4.4
Ball hardness	5 bearings (3 balls each)	3.3.2	4.4.5.1
Ring hardness	5 bearings (all rings)	3.3.1	4.4.5.1
Ball geometric quality	2 bearings (all balls)	3.4.5	4.4.1.1.1
Compatibility and corrosion resistance	2 bearings (2 balls each, 2 rings each, and 1 cage each)	3.2.1	4.4.7

NOTES:

1/ The examination and tests shall be performed in the order listed in this table.

2/ Certification sheets from the material suppliers that include test methods and results may be used to verify the material properties instead of the verification tests.

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TABLE VII. Classification of defects.

Category	Defect
Critical:	
001	Bearing vibration exceeds allowable limits (see table V and 3.6)
002	Bearing fails fatigue life test (see 3.5) 1/
Major:	
101	Bore diameter is not within required limit (see 3.4)
102	Outside diameter is not within required limit (see 3.4)
103	Width of individual ring is not within required limit (see 3.4)
104	Radial internal clearance is not within required limit (see 3.4)
105	Seal or shield inhibits the free rotation of bearing (see 3.4.7)
106	Hardness of each ring is not within required limit (see 3.3.1)
107	Hardness of each ball is not within required limited (see 3.3.2) 1/
108	Balls are not within required limits (see 3.4.4)
109	Bearing lubricant or preservative is missing or incorrect lubricant or preservative is used (see 3.7)
110	Radial runout of inner or outer rings is not within required limit (see 3.4) 1/
111	Bearings components are not new and unused (see 3.4)
112	Individual bearing shows evidence of corrosion (see 3.10)
113	Individual bearing shows evidence of contamination (see 3.10)
114	Negative stickout or preload of bearing is not within required limit (see 3.4.9)
115	Duplex pair is not universal ground (see 3.4.3)
116	Cage material is not compatible with bearing components and lubricant, and use prevents bearing from meeting performance requirements (see 3.2.5 and 3.4.5)
117	Inner ring reference face runout with bore is not within the required limit (see 3.4.7)
118	Axial runout of assembled bearing inner ring is not within the required limit (see 3.4.7)
119	Outside cylindrical surface runout with outer ring reference face is not within the required limit (see 3.4.7)
120	Axial runout of assembled bearing outer ring is not within the required limit (see 3.4.7)
121	Chamfer dimensions are not within required limits (see 3.4.7)
122	Functional surface of inner and outer rings and balls do not meet requirements (see 3.10)
123	Bearing is not packaged as required (see 5.1)
124	Ring materials do not meet requirements (see 3.2.3)
125	Ball materials do not meet requirements (see 3.2.4)
126	Cage, shield, or seal materials do not meet requirements (see 3.2.5 and 3.2.6)

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TABLE VII. Classification of defects – Continued.

Category	Defect
Minor:	
201	Nonfunctional surface of inner and outer rings and balls do not meet requirements (see 3.10)
202	Bearing is not marked as required (see 3.8)
NOTE: 1/ Not performed during the conformance inspection of bearing lots.	

4.2.1 Waiver of first article inspection. The Technical Authority may waive either portions of or the entire first article inspection and test after the bearing vendor demonstrates to the Technical Authority's satisfaction that the bearings offered for delivery meet the first article requirements.

4.3 Conformance inspection. Before a lot of bearings is accepted, the conformance inspection and tests specified in table VIII shall be performed on material samples and a sample of bearings selected from the lot. Unless otherwise specified (see 6.2), the material and bearing samples shall be evaluated for the defects listed in table VII, except as noted therein. Unless otherwise directed by the Technical Authority, the selected sample of assembled bearings shall be sent to the Naval Surface Warfare Center, Carderock Division, Bethesda, MD for conformance inspection prior to acceptance.

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TABLE VIII. Conformance inspection.

Examination or test	Test specimen	Performance requirement	Verification procedure
Properties of ring and ball materials 1/	Samples of ring and ball material	3.2.3 3.2.4	4.4.6.1 4.4.6.2
Ball geometry	1/	3.4.5	4.4.1.2.1
Ring hardness	--	3.3.1	4.4.5.2
Bearing construction and metrology	2/	3.4	4.4.1.2
Vibration	2/	3.6	4.4.3
Workmanship	2/	3.10	4.4.2
Lubricant	2/	3.7	4.3.3
Marking	2/	3.8	4.4.2

NOTE:

1/ Certification sheets from the material suppliers that include test methods and results may be used to verify the material and properties instead of the verification tests.

2/ The number of finished bearings in the sample shall be selected in accordance with 4.3.1.2.

4.3.1 Conformance inspection sampling.

4.3.1.1 Ball sampling. The balls shall be examined before assembly into the bearings. The number of balls in the sample shall be determined by the total number of balls used in the bearing in accordance with table IX.

TABLE IX. Ball sample size for conformance inspection.

Total number of balls used in the lot of finished bearings	Sample size
Up to 170	5
171 to 288	6
289 to 544	8
545 to 960	10
961 and larger	12

NOTE: The sample size is based on a zero-acceptance number sampling in accordance with MIL-STD-1916, level I.

4.3.1.2 Bearing sampling. The number of finished bearings in the sample shall be selected in accordance with table X.

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TABLE X. Bearing sample size for conformance inspection.

Lot size [1/]	Sample size [2/]
Up to 12	All
13 to 288	12
289 to 544	16
545 to 960	20
961 to 1,632	24
1,633 and larger	32

NOTES:

[1/] Duplex pairs shall be counted as one bearing.

[2/] The sample size is based on a zero-acceptance number sampling in accordance with MIL-STD-1916, level II.

4.3.2 Lot. A lot shall consist of assembled bearings of a particular standard part number, assembled at the same time on the same assembly setup, and processed through all final assembly operations as a single group. Each bearing of a lot shall be assigned a serial number, which shall be traceable to the materials used to manufacture the bearing and its assembly sequence. The lot records shall be maintained for no less than 15 years.

4.3.3 Disposition of lot sample. Lot samples that pass the conformance inspection shall be handled as follows. Greased bearings shall be re-lubricated and included in the lot for delivery. Oil-lubricated bearings shall be cleaned and re-lubricated and included in the lot for delivery. The standard part number and the marking “NT5” shall be obliterated from bearings that do not pass the conformance requirements.

4.4 Verification methods.

4.4.1 Metrology.

4.4.1.1 Metrology for first article inspection. All bearings subjected to the first article inspection shall be examined and measured to determine conformance with the requirements specified in 3.4.8. Runout measurements shall be made in accordance with ABMA 4.

4.4.1.1.1 Ball measurements. During the first article inspection, three balls from each of five disassembled bearings shall be measured in accordance with ABMA/ISO 10A and ASTM F2215 to verify conformance with the ball grade requirements specified by the bearing manufacturer (see 3.4.5). The diameter of all balls from two disassembled bearings shall be measured to determine conformance to the geometric quality requirements of ABMA/ISO 10A and ASTM F2215.

4.4.1.2 Metrology for conformance inspection. Metrological measurements shall be made on the lot sample of finished bearings. Measurements shall be made on the bearing bores, outside diameters, widths, radial internal clearances, and chamfer dimensions to determine conformance with 3.4.8. No measurements shall be made that require disassembly of the bearings. Additional metrology measurements may be made at the option of the Technical Authority.

4.4.1.2.1 Ball measurements. Measurements shall be made on the balls to be used in the bearings before they are assembled. The measurements shall be in accordance with ABMA/ISO 10A and ASTM F2215 to verify conformance with the ball grade requirements specified by the bearing manufacturer (see 3.4.5) and the geometric quality requirements of ABMA/ISO 10A and ASTM F2215.

4.4.1.2.2 Bore and outside diameter measurements. At a minimum, when performing bearing bore and outer ring diameter checks, a series of six individual readings shall be measured. The measurements shall consist of readings at a zero reference point and at 90 degrees from the zero reference point in three different planes. The zero position shall be located at a convenient marking (e.g., bed face dot) and then at 90 degrees from the selected marking. The top plane shall be located approximately 0.125 inch below the upper outer ring corner. The middle plane shall be located at the center of the bearing width and the bottom plane shall be located approximately 0.125 inch above the lower outer ring corner. Similarly, six measurements shall be performed on the bearing bore. The measurements shall be in accordance with ABMA 4.

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4.4.1.2.3 Radial internal clearance measurements. Radial internal clearance measurements shall be conducted in accordance with ABMA 4. The radial loads listed in table XI shall be used to conduct the radial play measurements as a function of outer ring diameter. Radial internal clearance measurements shall be conducted with the operating lubricant and the seals and shields shall not be removed. The measurement shall be made with the displacement indicator located centered over a ball whenever possible for consistency.

TABLE XI. Allowable loads for radial internal clearance measurements.

Bearing outer ring diameter (inches)	Radial load (pounds)
Less than or equal to 1.1811	0.5 – 5.5
1.3780 to 3.5433	5.0 – 11.0
3.9370 to 7.0866	11.0 – 33.0
7.4803 to 11.4173	22.0 – 44.0

4.4.2 Workmanship. The bearings shall be visually examined to verify conformance with the requirements for surface quality and workmanship (see 3.10) and marking (see 3.8). Irregularities may be observed under magnification to identify the defect. The defects shall be classified in accordance with table VII.

4.4.3 Vibration test. The bearings' vibration level shall be measured on the Bearing Analysis Machine (see 6.3). The measurements shall be made in accordance with ABMA/ISO 15242 and as indicated herein.

4.4.3.1 Physical quantity measured. The anderon shall be the physical quantity measured during the vibration test to determine compliance with the requirements of 3.6. The anderon is the radial vibrational velocity, measured on the outside diameter of the stationary outer bearing ring, with the inner ring rotating at 1,800 revolutions per minute, divided by the constant 297, and in units of microinches per second, root mean square. The anderon level shall be measured at a random location in the plane containing the centers of curvature of the ball groove to which the thrust load is applied during the vibration test.

4.4.3.2 Frequency ranges. The bearings' anderon levels shall be measured in three frequency ranges:

a. Low: 50 to 300 hertz (Hz)
b. Middle: 300 to 1,800 Hz
c. High: 1,800 to 10,000 Hz

4.4.3.3 Test lubricant. The sample bearings shall be shipped with the preservative lubricant in place. Grease-lubricated bearings shall be run in and the excess grease allowed to purge. Oil, in accordance with MIL-PRF-17331 (2190 TEP), shall be added to oil-lubricated bearings and the excess allowed to purge (see tables I, II, and III for required operating lubricant).

4.4.3.4 Test load. During the vibration test, an external thrust load shall be applied to the outer bearing ring in the direction(s) in which the bearing can accept operational thrust. The magnitude of the external thrust load shall be selected in accordance with table XII.

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TABLE XII. Thrust loads applied to the bearing during vibration testing.

Bore code	Thrust load on bearing (pounds)						
	Deep groove bearings: types 111 and 120 [1/]				Angular contact bearings: types 131, 133, and 134		
	100	200	300	400	7,200	7,300	7,400
00	--	20	20	--	--	--	--
01	20	20	20	--	--	--	--
02	--	20	20	--	--	--	--
03	--	20	30	--	60	60	--
04	--	20	35	--	70	70	--
05	--	20	45	--	90	90	--
06	--	30	55	75	110	110	--
07	--	45	65	--	130	140	205
08	--	50	75	--	150	175	260
09	50	55	85	--	170	245	--
10	--	60	100	120	200	280	380
11	--	80	115	--	230	330	--
12	--	95	135	--	270	390	560
13	--	105	150	--	300	450	--
14	--	120	170	--	350	510	--
15	--	130	195	--	400	580	--
16	--	140	215	--	470	660	--
17	--	155	240	--	530	750	--
18	--	190	270	--	580	935	--
19	--	220	295	--	--	1,040	--
20	--	250	355	--	--	--	--
21	--	280	385	--	775	--	--
22	--	315	455	--	--	1,490	--
24	--	360	455	--	--	--	--
26	--	410	525	--	--	--	--
32	--	550	--	--	--	--	--

NOTE:

[1/] The S3500 and S3600 series bearings in table II correspond to the 200 and 300 series bearings in this table, respectively.

4.4.3.5 System calibration. The mechanical and electrical frequency response of the instrumentation used with the bearing analysis machine shall be in accordance with ABMA/ISO 15242.

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4.4.3.6 Vibration measurements. The vibration level shall be measured at one random location on the periphery of the stationary outer ring for each direction in which the bearing can accept operational thrust load. Therefore, bearing types 111 and 120 require two vibration measurements per bearing, and types 131, 133, and 134 require one vibration measurement per bearing. The highest anderon levels measured in each frequency range (low, middle, and high) and at each location shall be recorded after the bearing has been run at the test speed and load for 5 minutes. For bearings that can be loaded in two directions, the highest readings measured in both load directions shall be used to determine conformance with 3.6. If a bearing does not meet the requirements in 3.6, the vibration transducer shall not be relocated to measure a lower vibration level.

4.4.3.7 Temperature measurements. The temperature rise at the bearing outer ring shall be monitored and recorded during the test. For normal steady-state operation, the temperature of the bearing outer ring, as measured on its outside diameter, shall not exceed 194 °F (90 °C).

4.4.4 Test for fatigue life and radial dynamic load rating.

4.4.4.1 Test procedure. A fatigue life test shall be conducted during the first article inspection to verify that a sample of assembled bearings, part number M17931-G061, meets the required dynamic load rating (see table I). The test bearing shall be interference-mounted on a shaft (see A.3.6) as shown on figure 1. A radial load of 5,000 pounds shall be applied to the bearing while it is rotated at 2,000 revolutions per minute for 1200 hours. The bearings shall be lubricated with 2190 TEP oil in accordance with MIL-PRF-17331; the oil lubrication system shall maintain an oil inlet temperature of 110±5 °F (43.5±2 °C).

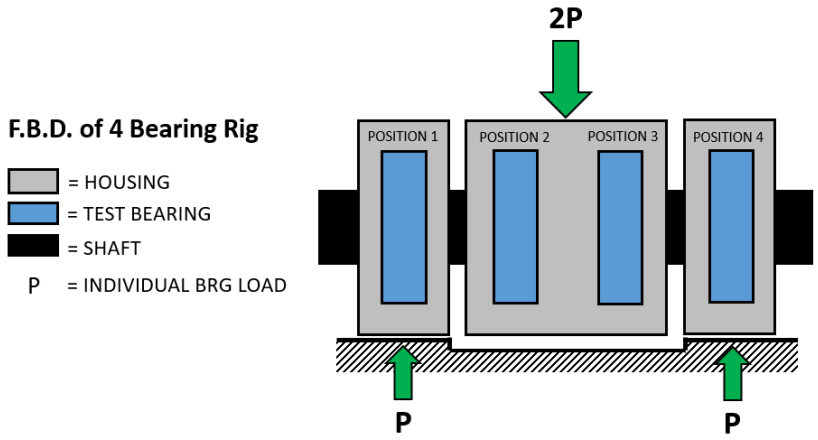


FIGURE 1. Bearing load-life test schematic diagram.

4.4.4.2 Temperature measurement. The temperature of the outer bearing ring shall be measured during the test at no less than every 24 hours, with the first measurement taken within one hour of initiating the fatigue test. The test shall be terminated if the bearing outer ring temperature exceeds 300 °F (148.9 °C).

4.4.4.3 Vibration test. Upon completion of the fatigue test, five bearings shall be selected at random, and these bearings shall be subjected to vibration testing in accordance with the verification method specified in 4.4.3.

4.4.4.4 Post-test inspection. After the vibration test, the tested bearings shall be disassembled and examined to determine the condition of the bearing parts and inspect for mechanical damage.

4.4.5 Hardness measurements.

4.4.5.1 Hardness measurement during the first article inspection. The hardness of both rings and three balls from each of five disassembled bearings shall be measured in accordance with ASTM E18 for the rings and ABMA/ISO 10A and ASTM F2215 for the balls. The defects shall be classified as specified in table VII. Balls subjected to the hardness test shall not be reused.

4.4.5.2 Hardness measurement during the conformance inspection. The hardness of both rings shall be measured in accordance with ASTM E18. The defects shall be classified as specified in table VII.

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4.4.6 Material properties.

4.4.6.1 Properties of ring material. The macrostructure, microstructure, inclusion content, and decarburization of the ring material shall be determined in accordance with SAE-AMS6444, ASTM A295/A295M, and ASTM E45, as applicable, to confirm that the material meets the requirements in 3.2.3.

4.4.6.2 Properties of ball material. The macrostructure, microstructure, inclusion content, and decarburization of the ball material shall be determined in accordance with ASTM A295/A295M and ASTM E45. The grain size of the ball material shall be determined in accordance with ASTM E112 to confirm that the material meets the requirements specified in 3.2.4.

4.4.7 Compatibility and corrosion resistance. The compatibility and corrosion resistance of the balls, rings, and cages shall be tested by the following procedure:

a. Immerse the components in the lubricating oil or grease that will be used with the bearing for 1 hour at 100±5 °F (38±3 °C), followed by air drying at the same temperature for one hour. Repeat this immersion and drying cycle 12 times.

b. At the completion of the test, examine the components for corrosion at 10× magnification.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The ball bearings covered by this specification are high precision and must meet the vibration requirements specified herein. The bearings are intended for use in low-vibration machinery. Current stock of quiet bearings may be used until depleted; future requisitions for quiet bearings should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a certificate of conformance is required for each lot.
c. Standard part number (see 1.2).
d. When first article inspection is required (see 3.1 and 4.2).
e. Bearing requirements (see 3.4).
f. Whether the required basic dynamic radial load rating (C_r) is different from that listed in tables I, II, or III (see 3.5).
g. The acceptance criteria for the conformance inspection (see 4.3).
h. Additional metrological measurements (see 4.4.2).
i. Packaging requirements (see 5.1).
j. Bearing vacuum seal packaging (see 6.3).
k. Packaging of duplex pairs (see 6.4).
l. When lubrication is other than specified (see A.3.2).
m. Applications in which bearings are not secured against shoulders by locknuts (see A.3.4).
n. Bearing fit-up for special design applications (see A.3.6).

6.3 Vacuum seal packaging. When specified (see 6.2), bearings fitted with single or double elastomer contact seals are not to be subjected to vacuum during the bag sealing process.

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6.4 Packaging of duplex pairs. When specified (see 6.2), packaging of duplex pairs will be as follows: Type 131, class 2; type 133, class 2; and type 134, class 2 bearings will be matched by the manufacturer as pairs. Final packaging of the bearing pairs will be in accordance with MIL-DTL-197, method 40B. The matched bearing pairs will be packaged together as a unit using the following procedure. Each bearing comprising the pair will be vacuum-formed plastic skin packaged individually. The individual bearings will then be stacked such that the markings of both bearings are visible. A final layer of plastic film will be vacuum formed over the pair and issued together as a unit. All seams will be sealed using acetone containing acetate material. All required labels (boxes and the outer layer of the bearing vacuum-formed skin) will be marked with the following caution statement: "MATCHED SET, DO NOT SEPARATE".

6.5 Bearing analysis machine. The bearing analysis machine is a commercially- or Navy-designed and constructed test device, utilizing a commercially available Anderometer Amplimeter and a mechanical drive/loading system mounted on a 2,000-pound seismic mass. The test bearing is mounted on a true-running spindle with flexible bearing seat and held in position by a thrust load transmitted through a cylindrical load ring. The bearing vibration is determined on the three-frequency-band vibration analyzer of the Anderometer Amplimeter.

6.6 Definitions. The following definitions are provided to establish the intended meaning of terms used in this specification.

6.6.1 Contact seal. A contact seal is a bearing seal that contacts one bearing ring with a predetermined pressure (see 3.5).

6.6.2 Functional surfaces. Functional surfaces are all ball and raceway surfaces (see 3.4 and 3.10).

6.6.3 Mean hardness. Mean hardness is the arithmetic mean of hardness measurements, recorded to the nearest tenth (see 3.3.2):

Mean hardness: $R = (R1 + R2+\ldots Rn)/n$, where R1…Rn are the hardness values of n samples in the lot.

6.6.4 Negative stickout duplex pair. A negative stickout duplex pair has no internal loading, independent of external radial or axial loads carried by the bearing. A negative stickout duplex pair is manufactured by relieving selected faces of the single bearing ring widths by a certain amount (negative stickout), creating a non-preloaded (free-running) condition when the pair is mounted (see 3.4.9).

6.6.5 Nonfunctional surfaces. Nonfunctional surfaces are all bearing surfaces not listed as functional surfaces (see 3.11).

6.6.6 Preloaded duplex pair. A preloaded duplex pair has an internal loading, independent of external radial or axial loads carried by the bearing. A preloaded duplex pair is manufactured by relieving selected faces of the single bearing ring widths by a certain amount (positive stickout), creating an internal preload when the pair is mounted (see 3.4.9).

6.6.7 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.7 Surplus bearings. Surplus bearings may be offered for acquisition, provided that:

a. The bearings are submitted for first article and conformance inspections as specified herein. The Technical Authority may waive the first article inspection if the bearing manufacturer has previously satisfied these requirements.

b. The contractor furnishes the Technical Authority with the Government contract number under which the bearings were fabricated.

6.8 Subject term (key word) listing.

Cage
Chromium – alloy, steel
Noise-tested
Race
Rings
Vibration

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6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPLICATION REQUIREMENTS FOR QUIET BALL BEARINGS

A.1 SCOPE

A.1.1 Scope. This appendix is a mandatory part of the specification for equipment that houses quiet ball bearings. The information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this appendix to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service

MIL-PRF-24508 - Grease, High Performance, Multipurpose

DEPARTMENT OF DEFENSE STANDARDS

MS19070 - Washers, Key Retaining, Ball and Roller Bearing, Regular Series

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

A.2.2 Non-government publications. The following documents form a part of this appendix to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA)

ANSI/ABMA 9 - Loading Rating and Fatigue Life for Ball Bearings

(Copies of this document are available online at www.americanbearings.org/.)

A.3 APPLICATION REQUIREMENTS

A.3.1 Bearing fatigue life. The fatigue life of single bearings shall be calculated in accordance with ANSI/ABMA 9. Bearing life adjustment factors shall not be used in the calculation unless specifically permitted by the equipment specification. The fatigue life of duplex bearings shall be calculated in accordance with ANSI/ABMA 9, with the following exceptions:

a. Duplex bearings shall be treated as two individual single-row bearings.

b. The user shall account for axial and radial forces from all external and internal loads, including centrifugal force, preload, shaft-to-bearing interference fit, and differential thermal expansion of the races and balls as the bearing heats up to operating temperature.

c. The user shall account for the reduced reliability of a two-bearing arrangement with the Palmgren combined life expectancy formula:

$$1/L^{10/9} = 1/L_1^{10/9} + 1/L_2^{10/9}$$

Where L_1 and L_2 are the calculated lives (L_{10}) of the individual bearings and L is the L_{10} life of the combined bearing.

A.3.2 Lubrication. Unless otherwise specified (see 6.2), bearings shall be lubricated with grease in accordance with MIL-PRF-24508. Grease-lubricated bearings shall be selected so that the product N*Dm is less than 350,000. The value N is the rotational speed of the rotating bearing ring in revolutions per minute. The value Dm is the mean diameter of the bearing, in millimeters; $Dm = \frac{(\text{bore diameter} + \text{outside diameter of the outer ring})}{2}$. If an N*Dm less than 350,000 cannot be achieved, or oil lubrication has a demonstrable advantage in a given application, then lubricating oil in accordance with MIL-PRF-17331 (2190 TEP) shall be used.

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A.3.3 Temperature. The equipment shall be designed so that quiet ball bearings operate at the lowest temperature possible. For normal steady-state operation, the temperature of the bearing outer ring, as measured on its outside diameter, shall not exceed 194 °F (90 °C). The temperature limitations of the cage and closure materials establish the maximum permissible steady-state operating temperature at 230 °F (110 °C). Intermittent temperature excursions to 300 °F (149 °C) are acceptable if the equipment designer demonstrates to the Technical Authority's satisfaction that the intermittent temperature excursions are not detrimental to the bearing.

A.3.4 Locknuts. Unless otherwise specified (see 6.2), a mounted bearing shall be secured against a shaft shoulder by a locknut/lockwasher combination or by a self-locking locknut. The locknuts and lockwashers may be purchased in accordance with MS19068 and MS19070, or the equivalent commercial grade. (The appropriate MS19068 and MS19070 size number is the same as the bearing's bore code.) In order to develop the full strength of a standard locknut, or to develop the full locking effect of a self-locking locknut, the nut shall fully engage the male threads. In all cases, when installed and tightened, protrusion of the male threads shall be flush with the top of the nut, at a minimum.

A.3.4.1 Self-locking locknuts. Self-locking locknuts (or prevailing torque retaining nuts) may be used instead of a conventional locknut/lockwasher combination and provide better locking action and more protection against loosening, especially from vibration. Self-locking locknuts are essentially the same as MS19068 locknuts; however, they have a nylon compression collar instead of a lockwasher, which has a 300 °F temperature limit.

A.3.5 Locknut torques. Standard locknut torques shall be used, regardless of the type of locknut. The bearing locknuts shall be tightened within the allowable locknut torque range for dry thread engagement as shown in table A-I. The locknuts shall not be hammered or beat upon.

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TABLE A-I. Allowable range of locknut torques.

Bearing bore code	Locknut torque (foot-pounds)
00	10-20
01	10-20
02	10-20
03	10-20
04	12-35
05	23-50
06	32-60
07	39-70
08	50-80
09	64-90
10	67-100
11	82-125
12	99-150
13	131-175
14	152-200
15	173-250
16	197-275
17	222-325
18	248-375
19	277-425
20	345-475
21	380-550
22	380-550
24	380-550
26	380-550
32	380-550

A.3.6 Bearing fit-up. The size and physical condition of the shaft seat on which a quiet bearing is mounted and the housing in which a quiet bearing is mounted shall be controlled. Achieving the correct fit is the most important factor in the proper mounting of noise-quiet bearings. Tables A-II and A-III give the required fits for the bearing inner ring bore-to-shaft seat and the bearing outer ring outside-diameter-to-housing-bore.

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TABLE A-II. Bearing-to-shaft fits (inner ring rotating).

Bearing bore code	Bearing bore diameter (inches)		Shaft seat diameter (inches)		Interference fit (inches)	
	Max	Min	Max	Min	Max	Min
00	0.3937	0.3935	0.3940	0.3938	0.0005	0.0001
01	0.4724	0.4722	0.4727	0.4725	0.0005	0.0001
02	0.5906	0.5904	0.5909	0.5907	0.0005	0.0001
03	0.6693	0.6691	0.6696	0.6694	0.0005	0.0001
04	0.7874	0.7872	0.7877	0.7875	0.0005	0.0001
05	0.9843	0.9841	0.9846	0.9844	0.0005	0.0001
06	1.1811	1.1809	1.1814	1.1812	0.0005	0.0001
07	1.3780	1.3778	1.3783	1.3781	0.0005	0.0001
08	1.5748	1.5746	1.5751	1.5749	0.0005	0.0001
09	1.7717	1.7715	1.7720	1.7718	0.0005	0.0001
10	1.9685	1.9683	1.9688	1.9686	0.0005	0.0001
11	2.1654	2.1652	2.1657	2.1655	0.0005	0.0001
12	2.3622	2.3620	2.3625	2.3623	0.0005	0.0001
13	2.5591	2.5589	2.5594	2.5592	0.0005	0.0001
14	2.7559	2.7557	2.7562	2.7560	0.0005	0.0001
15	2.9528	2.9526	2.9531	2.9529	0.0005	0.0001
16	3.1496	3.1493	3.1500	3.1497	0.0007	0.0001
17	3.3465	3.3462	3.3469	3.3466	0.0007	0.0001
18	3.5433	3.5430	3.5437	3.5434	0.0007	0.0001
19	3.7402	3.7399	3.7406	3.7403	0.0007	0.0001
20	3.9370	3.9367	3.9374	3.9371	0.0007	0.0001
21	4.1339	4.1336	4.1343	4.1340	0.0007	0.0001
22	4.3307	4.3304	4.3311	4.3308	0.0007	0.0001
24	4.7244	4.7241	4.7248	4.7245	0.0007	0.0001
26	5.1181	5.1177	5.1186	5.1182	0.0009	0.0001
32	6.2992	6.2988	6.2997	6.2993	0.0009	0.0001

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TABLE A-III. Bearing-to-housing fits (outer ring stationary).

Bearing OD (inches)		Housing bore diameter (inches)		Clearance fit (inches)	
Max	**Min**	**Max**	**Min**	**Max**	**Min**
1.1024	1.1022	1.1029	1.1026	0.0007	0.0002
1.1811	1.1809	1.1816	1.1813	0.0007	0.0002
1.3780	1.3778	1.3785	1.3782	0.0007	0.0002
1.4567	1.4565	1.4572	1.4569	0.0007	0.0002
1.5748	1.5746	1.5753	1.5750	0.0007	0.0002
1.6535	1.6533	1.6540	1.6537	0.0007	0.0002
1.8504	1.8502	1.8509	1.8506	0.0007	0.0002
2.0472	2.0470	2.0479	2.0475	0.0009	0.0003
2.4409	2.4407	2.4416	2.4412	0.0009	0.0003
2.8346	2.8344	2.8353	2.8349	0.0009	0.0003
2.9528	2.9526	2.9535	2.9531	0.0009	0.0003
3.1496	3.1494	3.1503	3.1499	0.0009	0.0003
3.3465	3.3463	3.3474	3.3469	0.0011	0.0004
3.5433	3.5431	3.5442	3.5437	0.0011	0.0004
3.9370	3.9368	3.9379	3.9374	0.0011	0.0004
4.3307	4.3305	4.3316	4.3311	0.0011	0.0004
4.7244	4.7242	4.7253	4.7248	0.0011	0.0004
5.1181	5.1179	5.1191	5.1186	0.0012	0.0005
5.5118	5.5116	5.5128	5.5123	0.0012	0.0005
5.9055	5.9053	5.9065	5.9060	0.0012	0.0005
6.2992	6.2990	6.3002	6.2997	0.0012	0.0005
6.6929	6.6926	6.6938	6.6934	0.0012	0.0005
7.0866	7.0863	7.0875	7.0871	0.0012	0.0005
7.4803	7.4800	7.4814	7.4809	0.0014	0.0006
7.8740	7.8737	7.8751	7.8746	0.0014	0.0006
8.4646	8.4643	8.4657	8.4652	0.0014	0.0006
8.8583	8.8580	8.8594	8.8589	0.0014	0.0006
9.4488	9.4485	9.4499	9.4494	0.0014	0.0006
10.2362	10.2359	10.2373	10.2368	0.0014	0.0006
11.0236	11.0233	11.0247	11.0242	0.0014	0.0006
11.4173	11.4170	11.4184	11.4179	0.0014	0.0006

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A.3.6.1 Variations of the bearing fit-up. Variations from the requirements of tables A-II and A-III may be appropriate in certain design applications. These variations (with detailed technical justification) shall be submitted for approval to the cognizant design review activity, Technical Authority, and the NAVSEA Bearing and Seal Technology Branch. Variations may be appropriate in applications such as the following:

a. Designs involving heavy radial loads on the bearing (as in some horizontal shaft machinery) may require a heavier interference fit of the bearing on the shaft. A heavier interference fit will normally require a bearing with a larger radial internal clearance (such as standard part numbers M17931-G083 and M17931-G087).

b. Designs with the bearing outer ring rotating and the inner ring stationary require a clearance fit at the bearing inner ring and an interference fit at the bearing outer ring.

A.3.6.2 Shaft seat and housing bore characteristics. The shaft seat and housing bore characteristics shall be as follows:

a. The tolerances for shaft seat and housing bore roundness shall be one-half the diametrical tolerances in tables A-II and A-III, respectively. The shaft seat roundness tolerance applies to diameter measurements made with a two-point gauge. If the shaft seat roundness is measured with a dial indicator with the shaft or housing on center, then the tolerances shall be one-quarter of those in tables A-II and A-III.

b. The minimum surface finish requirement is 32 roughness height rating (rhr) for all shaft seats and 63 rhr for all housings.

c. Shaft shoulders that contact the bearing ring side face shall be square to the shaft bearing seating surface within 0.0003 inch and shall not have an axial runout exceeding 0.0003 inch. Bearing housing and cap surfaces that contact the bearing ring side face shall be perpendicular to the axis and bearing seating surface of the housing bore within 0.001-inch total indicated runout.

A.3.7 Bearing radial internal clearance variations. The specified radial internal clearances for radial groove bearings (see tables I and II) are based on standard military applications. Variations from the specified clearances may be necessary in certain design applications, such as when typical operating conditions cause significant shaft heating and consequent closure of the bearing's internal clearance. Such variations (with detailed technical justification) shall be submitted for approval to the cognizant design review activity, Technical Authority, and the NAVSEA Bearing and Seal Technology Branch. If approved, procurement contracts should specify appropriate waivers (including revised marking requirements) to the requirements specified herein.

A.3.8 Installation requirements and guidance.

A.3.8.1 Measurement practice. The dimensions for shaft seats and housing bores shall be manufactured and measured to an accuracy of 0.0001 inch. A dial-indicating comparative snap gauge is recommended for measuring shaft diameters, and a dial-indicating bore gauge is recommended for measuring housing diameters.

A.3.8.2 Mounting practice. Quiet ball bearings are mounted with an interference fitted ring. Consequently, all mounting methods are based on obtaining the necessary interference fit without undue effort and without risking any damage to the bearing. In no case shall the force used to mount and dismount the bearings be transmitted through the balls. The mounting force shall be applied directly to the interference fitted ring. Therefore, the bearings shall be installed as follows:

a. In preparation for mounting, the bearings shall be heated for 1 hour at 203±10 °F (95±5.6 °C) in a thermostatically-controlled oven.

b. The locknuts shall be installed and torqued to the values given in table A-I while the bearings are hot. After the bearings cool to room temperature, the locknuts shall be loosened and re-torqued to the values of table A-1.

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A.4 SELECTION REQUIREMENTS

A.4.1 Bearing selection. New bearing designs shall be selected from the standard part numbers listed in table A-IV in order to limit, and eventually reduce, the number of bearings required for acquisition and stocking. Bearing designs other than those listed in table A-IV shall be approved by the Technical Authority. The proposed use of a bearing design not listed in table A-IV shall include the complete identification of bearing characteristics and the technical justification for its use.

TABLE A-IV. Quiet ball bearings approved for new equipment use.

Standard part number	Navy bearing code
M17931-A001	7203D
M17931-A002	7303D
M17931-A003	7204D
M17931-A004	7305D
M17931-A005	7306D
M17931-A006	7207D
M17931-A007	7207D
M17931-A008	7307D
M17931-A009	7407D
M17931-A010	7308D
M17931-A011	7308D
M17931-A011A	7308D
M17931-A012	7408D
M17931-A013	7209D
M17931-A014	7309D
M17931-A015	7309D
M17931-A016	7310
M17931-A017	7310D
M17931-A017A	7310D
M17931-A017B	7310D
M17931-A018	7310D
M17931-A019	7310D
M17931-A020	7310D
M17931-A021	7211D
M17931-A022	7311D
M17931-A023	7311D
M17931-A024	7312DB
M17931-A025	7312DF
M17931-A026	7312D
M17931-A027	7412D
M17931-A028	7313D

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TABLE A-IV. Quiet ball bearings approved for new equipment use – Continued.

Standard part number	Navy bearing code
M17931-A029	7313D
M17931-A030	7215D
M17931-A031	7315D
M17931-A032	7315D
M17931-A033	7217D
M17931-A034	7317D
M17931-A035	7317D
M17931-A036	318R
M17931-A037	318RD
M17931-A038	7318D
M17931-A039	7319D
M17931-A040	7319D
M17931-A041	7319D
M17931-A042	7319D
M17931-A043	7221D
M17931-A044	7322DB
M17931-A044A	7322DB
M17931-A045	7322DB
M17931-A045A	7322DB
M17931-C001	S3502
M17931-C002	S3503
M17931-C003	S3504
M17931-C004	S3604
M17931-C005	S3505
M17931-C006	S3605
M17931-C007	S3506
M17931-C008	S3606
M17931-C009	S3507
M17931-C010	S3607
M17931-C011	S3607
M17931-C012	S3608
M17931-C013	S3509
M17931-C014	S3609
M17931-C015	S3610
M17931-C016	S3610
M17931-C017	S3611
M17931-C018	S3612

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TABLE A-IV. Quiet ball bearings approved for new equipment use – Continued.

Standard part number	Navy bearing code
M17931-C019	S3613
M17931-C020	S3614
M17931-C021	S3615
M17931-C022	S3618
M17931-G001	200
M17931-G002	200SS
M17931-G003	101
M17931-G004	301Z
M17931-G005	202
M17931-G006	302Z
M17931-G007	203
M17931-G008	203Z
M17931-G009	203ZZ
M17931-G010	203SS
M17931-G011	303Z
M17931-G010	203SS
M17931-G011	303Z
M17931-G012	303SS
M17931-G013	204
M17931-G014	204Z
M17931-G015	204SS
M17931-G016	304
M17931-G017	304Z
M17931-G018	205
M17931-G019	205Z
M17931-G020	305
M17931-G021	305Z
M17931-G023	206
M17931-G024	206Z
M17931-G025	305
M17931-G026	306Z
M17931-G027	306ZZ
M17931-G028	306SS
M17931-G029	406Z
M17931-G030	207
M17931-G031	307
M17931-G032	307Z

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TABLE A-IV. Quiet ball bearings approved for new equipment use – Continued.

Standard part number	Navy bearing code
M17931-G033	307ZZ
M17931-G034	307SS
M17931-G035	208
M17931-G035A	208SS
M17931-G036	308
M17931-G037	308Z
M17931-G038	109SS
M17931-G039	209
M17931-G040	309
M17931-G041	309Z
M17931-G044	210
M17931-G045	310
M17931-G046	310Z
M17931-G047	310ZZ
M17931-G048	310SS
M17931-G049	410Z
M17931-G050	211
M17931-G051	311
M17931-G052	311Z
M17931-G053	311ZZ
M17931-G054	311SS
M17931-G055	212
M17931-G056	212Z
M17931-G057	312
M17931-G058	312Z
M17931-G059	312ZZ
M17931-G060	213ZZ
M17931-G061	313
M17931-G062	313Z
M17931-G063	313
M17931-G064	314
M17931-G065	314Z
M17931-G066	314ZZ
M17931-G067	215
M17931-G068	315
M17931-G069	315Z
M17931-G070	315Z

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TABLE A-IV. Quiet ball bearings approved for new equipment use – Continued.

Standard part number	Navy bearing code
M17931-G071	216ZZ
M17931-G072	316
M17931-G073	316Z
M17931-G074	217
M17931-G075	217ZZ
M17931-G076	317
M17931-G077	317Z
M17931-G078	318
M17931-G078A	219
M17931-G079	319
M17931-G080	319Z
M17931-G081	220Z
M17931-G082	320Z
M17931-G083	320
M17931-G084	321
M17931-G085	222
M17931-G086	322
M17931-G087	322
M17931-G088	232

A.4.2 Bearing identification. Equipment drawings and parts lists shall, as a minimum, identify MIL-DTL-17931 bearings by type and class, Navy code, and standard part number

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CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – SH
Air Force – 184
DLA – GS

Review activities:
Army – AT

Preparing activity:
Navy – SH
(Project 3110-2021-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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